Grade 4 General Science

Solved Exercises

Balochistan Textbook Board

Balochistan Textbook Board (BTBB)

Unit 1: Life Processes

Unit 1: Life Processes. Living things grow, take in nutrients, breathe, reproduce, remove waste, respond to their surroundings and die. Plants get energy from light through photosynthesis, while animals get energy from the food they eat.

1.1 Tick the correct option.

(i) The increase in the size of living things is called:

- Movement
- ✓ Growth
- Reproduction
- Excretion

(ii) The gas released by plants during photosynthesis is:

- Carbon dioxide
- Hydrogen
- ✓ Oxygen
- Nitrogen

(iii) During respiration animals take in:

- Nitrogen
- Hydrogen
- Carbon dioxide
- ✓ Oxygen

(iv) Blinking of eyes in response to bright light is an example of:

Respiration

✓ Sensitivity

Growth

Movement

(v) A place where plants and animals live and grow is called:

Air

✓ Habitat

Water

Light

(vi) Seeds are dispersed by:

Air

Water

Light

✓ All of these

1.2 Fill in the blanks.

(i) All living things need ____ to get energy.

food

(ii) Plants prepare their own food by the process of ____.

photosynthesis

(iii) Urine is formed in the ____.

kidney

(iv) Sweat is secreted through ____.

skin

(v) Majority of animals obtain their food directly from ____.

plants

(vi) Plants absorb water and nutrients through their ____.

roots

1.3 Match the column A with column B.

Match: Stomata, Lungs, Mimosa, Reproduction, Excretion, Birds.

Stomata: Small pores in plant leaves. Lungs: Organs of respiration. Mimosa: Closes its leaves when touched. Reproduction: Producing one's own kind of offspring. Excretion: Removal of waste products. Birds: Lay eggs.

1.4 Write short answers.

(i) What are basic characteristics of living things?

The basic characteristics of living things are growth, nutrition, respiration, sensitivity, reproduction and excretion. Plants and animals both show these life processes, with some similarities and differences between them.

(ii) How living things differ from non-living things?

Living things grow in size, take in food for energy, breathe, respond to changes around them, reproduce their own kind and remove waste from their bodies. Non-living things do none of these: they do not grow, eat, breathe, respond or reproduce.

(iii) What happens during the process of respiration?

During respiration, a living body exchanges gases with its surroundings to obtain energy from food. Animals take in oxygen from the air and release carbon dioxide; plants take in carbon dioxide in daytime and release oxygen, and at night they take in oxygen and release carbon dioxide.

(iv) Define sensitivity.

Sensitivity is the ability of living things to detect and respond to changes in their environment. For example, plants grow towards light, the leaves of the mimosa plant close when touched, and we blink when bright light shines into our eyes.

(v) What is habitat?

A habitat is a place where a plant or an animal lives, such as a forest for many plants, birds, insects and animals. A habitat provides everything a living organism needs, including food, water and space, and helps it to grow, reproduce and increase its population.

1.5 Write detailed answers.

(i) How do plants and animals reproduce?

Plants mostly produce flowers, in which fruits and seeds develop. When seeds mature, they are dispersed by wind, insects and animals, and under favourable conditions they germinate and grow into seedlings. In animals, reproduction happens in two ways: some females, like birds, fish, lizards and snakes, lay eggs which hatch after a set time, while others, like goats, cats, cows and humans,

give birth to their young ones.

(ii) Describe the importance of light and nutrients for plants growth.

Light is essential because plants use sunlight to prepare their own food by photosynthesis, and the food gives them energy for growth and other activities; plants grown in bright light are taller and healthier than plants kept in the dark. Nutrients are equally important: plants absorb nutrients such as nitrogen, phosphorus and potassium, along with minerals like calcium, magnesium and iron, from the soil through their roots, and these nutrients keep plants healthy and growing.

1.6 PROJECT

Project (hands-on classroom task, not a written answer): Germinate a few seeds of chick pea in a petri dish and observe the process of germination. Keep the seeds moist, watch daily, and record when the seed coat ruptures, the root comes out first, and then the shoot and leaves appear.

Unit 2: Plant Structure and Function

Unit 2: Plant Structure and Function. Plants are classified into flowering and non-flowering plants. A flowering plant has roots, stem, leaves, flowers and fruits, and each part has its own function; flowers carry out pollination, fruit and seed formation.

2.1 Tick the correct option.

(i) The male part of a flower is:

- Sepal
- Petal
- ✓ Stamen
- Carpel

(ii) The female part of a flower is:

- Sepal
- Petal
- Stamen
- ✓ Carpel

(iii) The part of the plant which absorbs water and nutrients is:

- ✓ Roots
- Stem
- Leaves
- Flowers

(iv) The process of photosynthesis takes place in:

Root

✓ Leaves

Seeds

Flowers

(v) The example of flowering plant is:

Quetta pine

Chalgoza pine

Junipers

✓ Rice

(vi) Flowering plants produce seeds in:

Cones

✓ Fruits

Leaves

Roots

2.2 Fill in the blanks.

(i) Flower is the ____ part of a plant.

reproductive

(ii) Mosses and ferns reproduce through ____.

spores

(iii) The underground part of a plant is called ____.

root

(iv) The colour of the leaves is ____.

green

(v) The stem provides ____ to the plant.

support

2.3 Match the column A with column B.

Match: Anther, Sepals, Petals, Chlorophyll, Ovary.

Anther: Swollen part of stamen in which pollen grains are produced. Sepals: Green leaves which cover a flower bud. Petals: Brightly coloured leaves of a flower. Chlorophyll: Green substance found in leaves. Ovary: Enlarged basal part of carpel.

2.4 Write short answers.

(i) Make a list of flowering plants.

Flowering plants include wheat, rice, corn, roses, sunflower, tomato, watermelon and grapes. Most plants on Earth today are flowering plants, and they produce flowers, then fruits which contain seeds.

(ii) How do non-flowering plants reproduce?

Non-flowering plants do not produce flowers or fruits. Mosses and ferns reproduce through spores, which are single-celled reproductive units dispersed by wind or water that develop into new plants, while conifers such as Quetta pine, juniper and chalgoza pine produce seeds inside cones.

(iii) How does transport of water take place in plants?

Plants absorb water through their roots: tiny root hairs absorb

water and dissolved minerals from the soil, the roots transport the water upward into the stem, and the stem moves it through small tubes to the leaves and other parts of the plant.

(iv) What are the three parts of a carpel?

The three parts of a carpel are the stigma, the style and the ovary. The stigma is the upper part which receives pollen, the style is the long tube-like structure connecting the stigma to the ovary, and the ovary is the enlarged basal part which contains the ovules.

(v) What is the function of stamens in a flower?

Stamens are the male reproductive part of a flower. Each stamen has an anther, the swollen top part where pollen grains containing male reproductive cells are formed, and a filament, the long thread-like structure that supports the anther, supplies it with water and nutrients, and helps in pollination.

2.5 Write detailed answers.

(i) Define life cycle. Draw the life cycle of a flowering plant.

A life cycle is the series of changes that a plant goes through during its life. A flowering plant starts from a seed, which germinates under favourable conditions of water, air and suitable temperature into a seedling; the seedling grows into a mature plant, which produces flowers, then fruits and seeds, and the mature seeds are dispersed so that a new life cycle begins again. Draw the cycle in order: seed, germination, seedling, mature plant with flowers, fruit, seeds, and dispersal back to a new seed.

(ii) What do the persons associated with following professions do?
(a) Botanist (b) Farmer (c) Florist (d) Gardener

(a) Botanist: studies plants, including their structure, function, reproduction, growth and uses. (b) Farmer: cultivates crops and looks after them, using tools and equipment for sowing, growing and harvesting, and also raises animals and looks after their health. (c) Florist: sells flowers and makes flower designs such as bouquets and wreaths, ensuring a supply of fresh flowers for weddings and other occasions. (d) Gardener: cultivates plants of interest on a small scale, improves landscapes by planting trees and flowers, and maintains lawns and controls weeds.

2.6 PROJECT

Project (hands-on classroom task, not a written answer): Draw a labelled diagram of a flower and explain the functions of sepals and petals. Sepals are the green leaves which cover a flower bud as it develops and protect its inner parts; petals are the brightly coloured leaves which protect the reproductive parts and attract pollinators like insects and birds by their colour and scent.

Unit 3: Animal Structure and Function

Unit 3: Animal Structure and Function. Animals are classified into vertebrates, which have a backbone, and invertebrates, which do not. Vertebrates fall into five classes: fishes, amphibians, reptiles, birds and mammals.

3.1 Tick the Correct option.

(i) All vertebrates have:

- Six legs
- Eight legs
- An exoskeleton
- ✓ A backbone

(ii) Which of the following is a flying mammal?

- Pigeon
- Eagle
- ✓ Bat
- Sparrow

(iii) Insects are a group of animals that have:

- Two legs
- Four legs
- ✓ Six legs
- Eight legs

(iv) Which of the following is not a part of digestive system?

Esophagus

Stomach

Large intestine

✓ Kidneys

(v) The front teeth in each jaw are called:

✓ Incisors

Canines

Premolars

Molars

(vi) Which of the following is not a fish?

Shark

Rohu

Goldfish

✓ Jellyfish

3.2 Fill in the blanks.

(i) Fishes breathe through ____.

gills

(ii) Insects body is divided into head, thorax and ____.

abdomen

(iii) An individual coral is called a ____.

polyp

(iv) Stomach is a sac-like ____ organ.

muscular

(v) The last part of human digestive system is ____.

anus

3.3 Match the column A with column B.

Match: Amphibians, Mammals, Reptiles, Insects, Teeth.

Amphibians: Frog and toad. Mammals: Cow, goat, rabbit etc. Reptiles: Lizard, snake, crocodile etc. Insects: Have six legs. Teeth: Chew food and break it into pieces.

3.4 Write short answers.

(i) Differentiate between vertebrates and invertebrates.

Vertebrates are animals that have a backbone in their body and a skeleton inside their body, for example fish, frog, lizard, pigeon, dog, cat and humans. Invertebrates are animals without a backbone, for example insects, spiders, worms, snails and jellyfish; most animals on Earth are invertebrates.

(ii) What are five classes of vertebrates?

The five classes of vertebrates are: (i) Fishes, (ii) Amphibians, (iii) Reptiles, (iv) Birds, and (v) Mammals.

(iii) How do fish breathe?

Fish breathe through gills. The gills absorb oxygen from the water, so fish can live and swim in water using their tail and fins.

(iv) Name any five reptiles.

Five reptiles are: lizard, snake, tortoise, crocodile and sea turtle. Reptiles have dry, scaly skin and a backbone; most live on land and lay eggs, and snakes have no legs and crawl on their abdomen.

(v) Why jellyfish is not considered a fish?

Jellyfish is not considered a fish because it does not have a backbone. It is an invertebrate with a jelly-like body and tentacles, found in the ocean, while true fishes are vertebrates with a backbone, fins and gills.

3.5 Write detailed answers of the following questions.

(i) Describe characteristic features of fishes and amphibians.

Fishes live in water, have a tail and fins which help them in swimming, breathe through gills that absorb oxygen from the water, and lay eggs in water; salmon, goldfish, rohu and shark are examples. Amphibians live on land but reproduce in water: frogs, toads and salamanders lay eggs in water, the larva comes out of each egg and breathes through gills like a fish, and when it becomes adult it breathes through lungs and skin and has four legs for moving on land.

(ii) Suggest measures to prevent tooth decay and gum diseases.

Tooth decay is caused when bacteria in the mouth produce acid which breaks down the enamel, the outer protective covering of teeth, making holes or cavities; gum diseases are also caused by bacteria, and infected gums become red, swollen and may bleed. To prevent them: brush your teeth regularly at least twice a day, eat a healthy and balanced diet, avoid eating too much chocolates,

candies and soda drinks, remove food particles and plaque on a daily basis, and have a regular dental examination so that problems are detected early.

3.6 PROJECT

Project (hands-on classroom task, not a written answer): Draw a labeled diagram of human digestive system and explain the function of stomach. The stomach is a sac-like muscular organ where food is churned and mixed with digestive juices before it passes on to the small intestine.

Unit 4: Ecosystem-adaptations

Unit 4: Ecosystem-adaptations. An ecosystem is made of living and non-living things in a place, and living things adapt to their surroundings: fish to water, birds to air, cactus to deserts, and pine and deodar to cold places.

4.1 Tick the correct option.

(i) The living component of an ecosystem is:

- Air
- Soil
- ✓ Plants
- Water

(ii) Any detectable change in the environment is called:

- ✓ Stimulus
- Response
- Microorganisms
- All of these

(iii) Which of the following is NOT correct about desert:

- Lack of water
- Has few plants
- Hot Weather
- ✓ Receives maximum rainfall

(iv) Adaptations in fish include:

Gills

Streamlined body

Swim bladder

✓ All of these

(v) The adaptations in birds include:

Wings and feathers

Light weight skeleton

Strong muscles

✓ All of these

(vi) The plant that grows in cold environment:

Cactus

Lotus

✓ Deodar

Algae

4.2 Fill in the blanks.

(i) A large area with many trees is called a ____.

forest

(ii) A water body smaller than a lake is called a ____.

pond

(iii) A flowing water body is called a ____.

river

(iv) Grassland has many different types of ____.

grasses

(v) Cactus is an example of ____.

desert plant

4.3 Match the column A with column B.

Match: Cactus, Frog, Sunflower, Mimosa plant, Needle-like leaves.

Cactus: Fleshy stem. Frog: Hibernates in winter. Sunflower: Turns its head to face the sun. Mimosa plant: Closes its leaves when someone touches it. Needle-like leaves: Pine.

4.4 Write short answers.

(i) Define ecosystem.

An ecosystem is an area where living and non-living things interact with each other. Forests, ponds, rivers, grasslands and deserts are all different types of ecosystems.

(ii) How forest differs from a grassland?

A forest is a large area with many trees, along with many small plants and animals like birds, insects, reptiles and mammals. A grassland has many different types of grasses but very few trees, and animals like deer, elephant, zebra, lions and birds are found there.

(iii) What is hibernation?

Hibernation, also known as winter sleep, is a survival strategy in which many animals become inactive during winter to conserve energy when there is a lack of food and water. They slow down

their heart beat rate and breathing and use stored fat as a source of energy; frogs, snakes, bats and bears hibernate during winter.

(iv) Why do animals camouflage?

Animals camouflage, that is, they blend in with their surroundings, so that they are less visible to their predators and can survive better. For example, the green colour of grasshoppers makes them less visible in the leaves of plants, chameleons change colour to match their surroundings, and the white fur of a polar bear helps it blend in with snow and ice.

(v) Which plants and animals are found in desert ecosystem.

Desert plants include cactus, which stores water in its small thick leaves and fleshy stem. Desert animals include camels, lizards, snakes and scorpions. Deserts get very little rain each year and have very few plants because of lack of water and high temperature.

4.5 Write detailed answers.

(i) How living things respond to environmental conditions? Explain with the help of examples.

All living things detect changes in their environment and respond to them, and these responses help them to find food, avoid danger and adapt to changing conditions. Any detectable change in the environment is called a stimulus, for example light, temperature, sound, taste, smell and touch. Plants grow towards sunlight and a sunflower turns its head to face the sun, birds migrate to avoid cold winters, a frog hibernates in winter, we blink in response to sharp

light, and a dog salivates at the smell of food.

(ii) Discuss the effects of population growth and urbanization on habitats.

Human population is increasing rapidly, and with it the need for food, water, clothing and housing has increased, so more roads, buildings, dams and vehicles are being built, and factories and vehicles burning fossil fuels cause pollution, all of which harm the habitats of living things. Urbanization, the movement of population from rural areas towards cities for better health, education and employment, expands cities and destroys natural habitats like forests and wetlands, causing loss of habitats, overcrowding, lack of sanitation and drinking water, and air pollution. When a habitat changes, the organisms living in it are affected too: cutting trees on a large scale causes deforestation, animals lose food and shelter, and are compelled to migrate or die and may become extinct forever.

4.6 PROJECT

Project (hands-on classroom task, not a written answer): Make a model of any ecosystem. Choose one ecosystem, for example a pond or a forest, and show its living things (plants and animals) together with its non-living parts (water, soil, air, light).

Unit 5: Matter and its Characteristics

Unit 5: Matter and its Characteristics. Matter is anything that has mass and occupies space, and it exists in three states: solid, liquid and gas. Metals are shiny, strong conductors of heat and electricity, while insulators do not allow electricity to pass through them.

5.1 Tick the correct option.

(i) The state of matter in which particles move freely is:

- Solid
- Liquid
- ✓ Gas
- All of these

(ii) In solids, the particles are:

- Widely spaced
- Freely moving
- ✓ Tightly packed
- Loosely packed

(iii) The good conductor of heat and electricity is:

- Wood
- Rubber
- Plastic
- ✓ Copper

(iv) Which of the following is true about liquids?

Have definite shape

indefinite volume

Can not flow

✓ Can flow easily

(v) The quantity of matter in an object is called:

Volume

✓ Mass

Density

Conductivity

(vi) Which of the following will float on the surface of water?

A coin

A needle

An eraser

✓ A ping pong ball

5.2 Fill in the blanks.

(i) Anything that has mass and occupies space is called ____.

matter

(ii) The three states of matter are; solid, liquid and ____.

gas

(iii) Solids have a definite shape and ____.

volume

(iv) Liquids have definite volume but indefinite ____.

shape

(v) Gases have indefinite shape and ____.

volume

5.3 Match the column A with column B.

Match: Aluminium, Insulators, Metals, Solids, Liquids.

Aluminium: A good conductor of electricity. Insulators: Do not allow electricity to pass through them. Metals: Have a shiny and lustrous appearance. Solids: Have definite shape. Liquids: Can flow easily.

5.4 Write short answers.

(i) What are the three states of matter?

The three states of matter are solid, liquid and gas. A solid has a definite shape and volume, a liquid has a definite volume but no definite shape of its own, and a gas has neither a definite shape nor a definite volume.

(ii) How solids differ from liquids?

Solids are hard, do not easily bend or change their shape, have a definite shape and volume because their particles are very close together and tightly packed, and cannot flow because their particles only vibrate around fixed positions. Liquids have a definite volume but no fixed shape: they take the shape of their container, can flow because their particles can move from one place to another, and their particle arrangement is not fixed.

(iii) Define volume and its unit.

Volume is the amount of space occupied by an object, and it deals with the length, width and height of an object. The unit of volume is cubic meter (m3), and volume is calculated as Length x Width x Height.

(iv) Why does a needle sinks in water?

A needle sinks in water because floating or sinking depends upon density, and a metal needle has a higher density than water. Objects float when they have less density than water, like ice cubes, and sink when they have higher density than water, like coins.

(v) What processes are associated with heating?

Heating is associated with melting and boiling: melting is the process by which a solid changes to a liquid state on heating, and boiling is the process by which a liquid changes into a gaseous state when it is heated. Further heating can vapourize a liquid into a gas; the melting point of ice is 0 degrees C and the boiling point of pure water is 100 degrees C.

5.5 Write detailed answers.

(i) Describe the characteristics of gases.

Gases do not have a definite shape and volume: they occupy all available space and take the shape of their container, and their particles move freely in all directions. Gases can be easily compressed because there are a lot of empty spaces between their molecules. Examples of gases include nitrogen, oxygen, carbon dioxide and hydrogen, and the air surrounding the Earth is a

mixture of different gases.

(ii) Describe some properties of metals.

Metals are elements which are shiny and good conductors of heat and electricity. Appearance: metals like gold, silver and copper have a shiny appearance and are widely used in jewellery making. Texture and colour: the texture of metals may be smooth, rough, soft, hard or bumpy, and most metals have a silvery-grey colour, though gold is deep yellow and copper is red-orange. Density: different metals have different densities; aluminium is a light metal with low density used in laptops and mobile phones, while iron is a heavy metal with high density used in buildings. Conduction: metals like silver, copper, stainless steel, iron and aluminium are good conductors of heat and electricity, so they are used in cooking utensils and in electrical wiring.

5.6 PROJECT

Project (hands-on classroom task, not a written answer): List down the metals, non metals, conductors and insulators used at your home. For example, copper wires and steel spoons are metals and conductors, while wood, plastic and rubber are insulators and non-metals.

Unit 6: Forms of Energy and Energy Transfer

Unit 6: Forms of Energy and Energy Transfer. Energy is the ability to do work, and it appears as light, sound, heat, electrical and magnetic energy. Light travels in a straight line, sound is produced by vibrations, and heat is measured by a thermometer.

6.1 Tick the correct option.

(i) Torch is source of:

- ✓ Light energy
- Sound energy
- Magnetic energy
- Electric energy

(ii) An anemometer measures:

- A speed of light
- ✓ Wind speed
- Loudness of sound
- Flow of electron

(iii) Formation of shadow proves which property of light?

- ✓ Light always travels in a straight line
- Light bends
- Light dissipates
- None of these

(iv) Atoms and molecules move to produce:

Magnetic energy

Electrical energy

Water energy

✓ Heat energy

(v) The sound produced by chirping of birds is:

Hard

✓ Soft

Loud

Noise

6.2 Fill in the blanks.

(i) Energy is the ability to do ____.

work

(ii) Energy through which we see objects around us is called ____.

light energy

(iii) Sun, stars, lightning falls under the ____ source of light.

natural

(iv) A thermometer measures ____.

temperature

(v) Sound is also a form of ____.

energy

(vi) Water freezes at ____ degrees C.

0

6.3 Match the column A with column B.

Match: Reflection, Thermometer, Electrical energy, Rainbow, Loud sound.

Reflection: Bouncing back of light. Thermometer: Temperature measurement. Electrical energy: Flow of electrons. Rainbow: Seven colours. Loud sound: Lightning.

6.4 Write short answers.

(i) Define energy, give any two examples.

Energy is defined as the ability of a body to do work, and we need energy to perform our daily works. Two examples are light energy, which allows us to see objects around us, and sound energy, which makes us able to hear through our ears.

(ii) How is sound produced?

Sound is produced when the source vibrates: the only cause to produce sound is vibration, the back and forth motion of an object. For example, when you pluck a guitar string it vibrates and makes sound, when you clap your hands your palms hit each other and vibrate, and when we talk the vocal cords in our throat vibrate to produce sound.

(iii) What is the difference between the soft and loud sound?

A sound is soft when the vibrations that make it are very gentle and carry less energy, and it produces a pleasant effect on the ears, for example whispering and chirping of birds. A sound is loud when the

vibrations produced by it are very strong, and it may leave an unpleasant effect on our ears, for example the horn of a big truck, thundering and shouting.

(iv) Differentiate between the natural and artificial sources of light.
Natural sources of light are those not made by humans: the Sun is the main natural source of light, and stars, lightning and fireflies are other natural sources. Artificial sources of light are those created by humans, and they include lantern, street light, LED lights, tube light, candle and torch.

(v) What is magnetic energy?
Magnetic energy is the energy of magnets that attracts other magnets and magnetic materials. Magnets have energy that attracts other magnets and magnetic materials, and this energy is called magnetic energy.

6.5 Write detailed answers.
(i) What is light energy? Also state the properties of light.
Light energy is a form of energy that allows us to see various objects around us; the Sun is the main natural source of light energy, and plants prepare their food in the presence of sunlight. Its properties are: light travels in a straight line, known as a light ray, and it does not bend or twist around but goes straight ahead unless something blocks it or changes its path; light spreads out in all directions with very high speed, and the speed of light in vacuum or air is faster than any other thing, so light from the Sun reaches Earth in about 8 minutes 20 seconds; and when light hits

an object it bounces back, which is called reflection of light, while a shadow is formed when an opaque object blocks straight light rays.

(ii) Define sound. How does the sound produce?

Sound is a type of energy produced by vibrations, which makes us able to hear, and it travels through a medium such as air or water in waves from the source to our ears. Sound is produced when the source vibrates: a plucked guitar string vibrates and makes sound, clapped palms hit each other and vibrate, a drum membrane vibrates upon hitting, and our vocal cords vibrate when we talk. The only cause to produce sound is vibration, and a vibrating ruler hanging off a desk also produces a sound that can be heard by bringing the ears close to it.

6.6 PROJECT

Project (hands-on classroom task, not a written answer): Take a clinical thermometer and observe body temperature of your friend. Note it down. A healthy body temperature is about 37 degrees C; record the reading you actually observe and compare it with the normal value.

Unit 7: Force and Simple Machines

Unit 7: Force and Simple Machines. Force is simply a push or a pull, and it can move, stop, change the speed, direction or shape of an object. Simple machines like levers, pulleys, gears and ramps make our work easier, simpler and faster.

7.1 Tick the correct option.

(i) A force is a:

- Type of energy
- ✓ Push or a pull
- Loud Sound
- Fast movement

(ii) When we kick a football, the force used is called:

- Gravitational force
- Resistance force
- ✓ Muscular force
- Magnetic force

(iii) When we walk, our shoes grip the ground due to:

- Gravity
- ✓ Friction
- Electric force
- Magnetic force

(iv) Friction is reduced by:

Making surface rough

✓ Using oil or lubricant

Increasing weight

Using tape

(v) Why does a book remain at rest on a table?

Gravitational force

✓ Support force of table

Electric force

Magnetic force

(vi) A simple machine that uses a wheel and a rope is called:

Lever

Ramp

✓ Pulley

Gear

7.2 Fill in the blanks.

(i) A force is a push or a ____.

pull

(ii) When a juice bottle is squeezed, the force changes its ____.

shape

(iii) The force that slows down objects moving through air or water is called the ____ force.

resistance

(iv) Opposite electrically charged bodies ____ each other.

attract

(v) Friction reduces the ____ of machines.

efficiency

7.3 Match the column A with column B.

Match: Ramp, Gear, Lubricant, Gravitational force, Simple machine.

Ramp: Inclined plane. Gear: Wheel with teeth. Lubricant: Reduces friction. Gravitational force: Force pulling everything towards itself. Simple machine: Tool that makes work easier.

7.4 Write short answers.

(i) Name two effects of force on a moving object.

Two effects of force on a moving object are: it can stop or tend to stop a moving object, for example a goal keeper stops the moving ball by catching it, and it can change the speed of the moving object, increasing or decreasing it, for example pushing the accelerator pedal increases a car's speed and pushing the brake pedal decreases it.

(ii) Differentiate between magnetic force and electric force.

Magnetic force is the force of attraction between a magnet and a magnetic material: a magnet sticks to an iron rod and attracts metals such as iron, nickel and cobalt. Electric force is the attraction or repulsion between two charged bodies: opposite electrically charged bodies attract each other, while like charges repel each other, for example two balloons rubbed with hair both

become negatively charged and repel each other.

(iii) Give one example of advantages and disadvantages of friction.

Advantage: friction helps us to walk and run on the ground, and we can write in a notebook and light a match stick because of friction.

Disadvantage: friction reduces the efficiency of machines, and the soles of our shoes tear and the tyres of vehicles tear apart due to friction.

(iv) Define lever with example.

A lever is a simple machine that consists of a bar and rotates about a fixed point called the fulcrum: the force is applied at one end and the weight is lifted at the other end, and a lever can also change the direction of force. Examples include a sea-saw, a bottle opener and a pair of scissors.

(v) Define ramp with its uses.

A ramp is a sloped plane, also called the inclined plane. Ramps help us to carry heavy loads to a height with less effort: they let us just push or pull something instead of lifting it straight up. Wheelchair ramps and sloping ramps are examples of ramps.

7.5 Write detailed answers.

(i) State uses of friction.

Friction has many uses: it helps us to walk and run on the ground; we can write in a notebook and on the whiteboard or blackboard because of friction; we become able to light a match stick because of friction; the brakes of vehicles, bikes and bicycles work because

of friction to slow down or stop; and knots stay tied because of friction. Friction is a kind of resistive force that arises between any two objects in contact, it reduces the speed of objects, and it always acts in the opposite direction of motion.

(ii) What is force and what effects it can produce?

Force is simply a push or a pull: whenever you push or pull something, you use force, for example pushing a door to open it, pulling a drawer, pushing a toy car, or a train engine pulling its wagons. Force is invisible but we can feel and observe the changes due to it, and it can produce these effects on an object: it can make or try to make a still object move, it can stop or tend to stop a moving object, it can change the speed of a moving object, it can change the direction of moving objects, and it can change the shape or size of an object, for example the shape and size of an empty juice box changes whenever you squeeze it.

7.6 PROJECT

Project (hands-on classroom task, not a written answer): List down daily life examples and uses of the following machines. (a) Ramp: wheelchair ramps and sloping ramps, used to carry heavy loads to a height with less effort. (b) Lever: sea-saw, bottle opener and a pair of scissors, used to lift a weight with force applied at the other end of a bar turning about a fulcrum. (c) Pulley: lifting a bucket from a well, elevators, cranes and flagpoles, used to pull up straight heavy loads with small force.

Unit 8: Technology in Everyday Life

Unit 8: Technology in Everyday Life. Technology is the use of scientific knowledge and tools to make life better and work easier, from computers and robots to a simple pencil, hammer or nail. Scientific instruments like the thermometer, blood pressure apparatus, digital balance, stop watch and calculator help us measure, calculate and observe things precisely.

8.1 Tick the correct option.

(i) Which of these is not an example of technology in everyday life?

- Mobile phone
- Calculator
- ✓ Stone in a river
- Stop watch

(ii) A device used to measure body temperature is called:

- ✓ Thermometer
- Calculator
- Stop watch
- Digital balance

(iii) Which 3D shape has 6 faces and 12 edges?

- ✓ Cube
- Sphere
- Cone
- Cylinder

(iv) Which of these objects is shaped like a cylinder?

A football

✓ A juice can

A brick

A hat

(v) Sphere has a number of flat surfaces:

✓ 0

1

2

3

(vi) A device used to preform mathematical calculations is:

Anemometer

Thermometer

Cylinder

✓ Calculator

8.2 Fill in the blanks.

(i) Technology makes our lives ____.

better

(ii) A ____ measures how cold or hot something is.

thermometer

(iii) A tool used to measure the time period accurately is called ____.

stop watch

(iv) A 3D shape that is perfectly round like a ball is called ____.

sphere

(v) A ____ helps us to do the math problems accurately.

calculator

8.3 Match the column A with column B.

Match: Technology, Digital balance, Plumb line.

Technology: Tools that make life easier. Digital balance: Measures weight accurately. Plumb line: Helps stand things straight vertically.

8.4 Write short answers.

(i) Define technology and give an example of technology used in everyday life.

Technology is the use of scientific knowledge and tools to make the life better and work easier, or any tool or invention created for the betterment of human lives. Examples range from computers, robots and complex machines to even a pencil, hammer and simple nail, and in everyday life a mobile phone, a calculator or a thermometer are all technology.

(ii) Write a short note on blood pressure apparatus.

Blood pressure apparatus is a simple tool used widely by nurses and doctors to measure the blood pressure: it tells how hard the blood is pushing against the walls of the vein in which blood is flowing, and it is an important health monitoring tool. Its main parts are the cuff, pump, gauge and the stethoscope. It has two types,

manual and digital: the manual type is used by wrapping the cuff around the arm, inflating it and listening to the blood pulse sound with the stethoscope, while the digital type inflates and deflates the cuff itself and gives the blood pressure and heart beat reading. The normal human blood pressure reading is 120 / 80 mmHg.

(iii) What do you know about the digital devices?

Nowadays there are wide varieties of available digital devices, and among these devices the mobile phones and tablets are master pieces of technology. They have a number of applications in them, such as stop watch, calculator, compass, camera and weather apps, and they are used for versatile purposes including communication, learning and knowledge.

(iv) Define plumb line and state its uses.

Plumb line is a very simple instrument with a piece of string and a small weight hanging with the string; the weight hanging with the string is called the plumb bob. When the plumb line is hanged down, it goes perfectly down in a straight vertical line and tells about the vertical straightness of something. Its use is to make things stand perfectly straight and vertical, for example to install a flagpole vertically: hold the plumb line at the top of the string, bring it close to the flagpole without touching it, and adjust the flagpole until the plumb line and flagpole become perfectly parallel.

(v) Enlist the items of first aid box.

Each first aid box contains: (i) Band aids or plasters, (ii) Antiseptic liquid or wipes, (iii) Cotton, (iv) Tweezers, (v) Painkillers, (vi) First

aid guide, (vii) Gauze pads, (viii) Adhesive tape, (ix) Pair of scissors, (x) Disposable gloves, and (xi) Antiseptic cream or ointment.

8.5 Write detailed answers.

(i) Write a short note on the following: (a) Thermometer (b) Digital balance (c) Stop watch.

(a) Thermometer: A device used to measure the temperature is called a thermometer; it actually tells us how hot or cold something is, and thermometers are of two types, general and clinical. It is widely used by doctors in hospitals to check if the patient has a fever or not, and it also tells us the temperature of our surroundings; the normal temperature of the human body is 37 degrees C or 98.6 degrees F. (b) Digital balance: A digital balance is a simple yet modern tool that measures the weight of objects very accurately, giving the weight in grams and kilograms, and it is widely used in shops and laboratories. Its use is very easy: just take the object you want to weigh, put it on the digital balance and it will give you the reading. (c) Stop watch: A stop watch is a simple tool used to measure time periods very accurately, with a simple stop and start button, and it helps us measure how fast an athlete can run or how long it takes to solve a math problem. Stop watches are of two types, manual and digital, both used with the start and stop button and a reset button; the manual stop watch has a clock dial and a needle that shows time, while digital stop watches show time in numbers, even telling the fractions of seconds.

(ii) Explain the plumb line in detail.

A plumb line is a very simple instrument with a piece of string and

a small weight hanging with the string, called the plumb bob. When the plumb line is hanged down, it goes perfectly down in a straight vertical line, and tells about the vertical straightness of something. You might have seen a flagpole standing perfectly straight and not leaning to one side: these are made perfectly straight with the help of a plumb line. To install a flagpole vertically using a plumb line: hold the plumb line at the top of the string and make sure the weight hangs freely vertically downward; carefully bring the plumb line close to the flagpole, making sure it does not touch the flagpole; look closely, and if the flagpole is not perfectly vertical, make it perfect by changing it in such a way that the plumb line and flagpole become perfectly parallel; then you have correctly installed the flagpole vertically.

8.6 PROJECT

Project (hands-on classroom task, not a written answer): Design a first aid box for your class or home. It must contain the items listed in 8.4(v): band aids, antiseptic liquid or wipes, cotton, tweezers, painkillers, a first aid guide, gauze pads, adhesive tape, a pair of scissors, disposable gloves and antiseptic cream or ointment, so that immediate help can be given for small cuts, scrapes or burns.

Unit 9: Earth and its Resources

Unit 9: Earth and its Resources. Natural resources are the materials found naturally on Earth which can be used by people, like sunlight, water, soil, air, trees, minerals, animals and fossil fuels. About 71% of Earth's surface is covered with water and 29% is land, and water keeps moving between the surface and the atmosphere in the water cycle.

9.1 Tick the correct option.

(i) Which of the following is not a natural resource of the earth?

- ✓ Cement
- Water
- Air
- Soil

(ii) How much Earth's surface is covered with water?

- 27%
- 29%
- ✓ 71%
- 50%

(iii) The process of evaporation involves the conversion of:

- Solid to liquid
- Liquid to solid
- ✓ Liquid to gas
- Gas to liquid

(iv) Icebergs are the broken pieces of:

Sand

Rock

Ice

✓ Glacier

(v) The conversion of water vapour into liquid droplets is called:

Evaporation

✓ Condensation

Snow

Rain

9.2 Fill in the blanks.

(i) Natural resources are found ____ on earth.

naturally

(ii) Air is a mixture of ____.

gases

(iii) Earth's surface is covered with about ____ percent water.

71

(iv) The broken pieces of glaciers are called ____.

icebergs

(v) The top most layer of Earth's surface is known as ____.

soil

9.3 Match the column A with column B.

Match: Water, Air, Iceberg, Forest, Soil.

Water: 3/4 of Earth's surface. Air: A mixture of gases. Iceberg: Broken piece of glacier. Forest: A natural resource. Soil: In which plant roots grow.

9.4 Write short answers.

(i) What are the natural resources of Earth?

Natural resources are the materials found naturally on Earth which can be used by people. Examples are sunlight, water (ocean, sea, river), soil, air, trees, minerals, animals and fossil fuels, and human beings use these natural resources for survival and upgrading living standards of life.

(ii) Define water bodies and their sources?

Water bodies are the places where water is found on Earth, in the form of oceans, lakes, rivers and underground water, and three fourth of Earth's surface is covered by water. The sources of water are: oceans and seas, which are the main sources of salt water, and rivers, lakes, springs, ponds, icebergs, glaciers and underground water, which are the sources of fresh water.

(iii) Define glaciers and icebergs?

Glaciers and icebergs are the frozen sources of fresh water on mountains. Glaciers are the large, massive masses of ice that are formed over the years, while the broken pieces of glaciers are icebergs; when temperature increases, glacier water flows

downward towards slopy areas and reaches streams, rivers, lakes and ponds.

(iv) How are the clouds formed?

Water in oceans, rivers and lakes warms up due to heat of the Sun and evaporates in the form of invisible gas called water vapour. This water vapour rises up and turns into liquid droplets or ice crystals, a process called condensation, and a large number of these tiny droplets or ice crystals together form clouds.

(v) Define water cycle.

The water cycle is the continuous movement of water from the Earth's surface to the atmosphere and then back to the ground. It involves different weather events like cloud formation, dew formation, the evaporation of puddles, snow and rain.

9.5 Write detailed answers.

(i) Describe at least 3 natural resources of Earth in detail.

Air: Air is a mixture of gases that surround our Earth, mostly made of nitrogen, oxygen and carbon dioxide; nitrogen is the most abundant gas in air, approximately 78%, that makes soil fertile for plant growth, oxygen is breathed by human beings and used by aquatic animals as dissolved oxygen, plants need carbon dioxide for photosynthesis, and oxygen in air is also used in the burning process. Water: Water is a vital natural resource for all living things; three fourth of Earth's surface is covered by water, but only 3% of Earth's total water is fresh water while 97% is salt water found in oceans and seas, and fresh water is found in land water bodies like

lakes, rivers, streams and glaciers. Soil and sunlight: Soil is the layer in which plant roots grow and crops and forests grow from it, and sunlight is a natural resource that warms the Earth, drives evaporation in the water cycle, and lets plants prepare their food.

(ii) Define fresh and salt water bodies.

Fresh water is water that contains a low amount of salt and is suitable for drinking: only 3% of Earth's total water is fresh water, and it is found in land water bodies like lakes, rivers, streams and glaciers, and in springs, ponds, icebergs and underground water; less than 1% of Earth's fresh water is easily accessible for human use, and most fresh water is frozen ice in glaciers or underground water. Salt water is water that contains a large amount of dissolved salt, like sodium chloride and other minerals: 97% of total water on Earth is salt water, found in huge water bodies like oceans and seas, and this water is not suitable for human consumption, though many plants and animals rely on it for survival.

9.6 PROJECT

Project (hands-on classroom task, not a written answer): Discuss some ways to save water, and share them with your community members as well. Ways to save water include turning off taps when not in use, fixing leaking taps, using a bucket instead of a running hose for washing, and reusing water where it is safe to do so, because only a small part of Earth's water is fresh and easily accessible.

Unit 10: Earth in the Solar System

Unit 10: Earth in the Solar System. Our solar system comprises the Sun, eight planets, moons, comets and asteroids, with the Sun at the center and every other celestial body revolving around it. Earth is the third nearest planet from the Sun and the only planet that provides the conditions suitable for life.

10.1. Tick the correct option.

(i) In our solar system, which object is exactly at the center?

- Earth
- Moon
- ✓ Sun
- Comets

(ii) Which phase of the moon looks like complete, bright circle in the sky?

- New moon
- ✓ Full moon
- Crescent moon
- Half moon

(iii) Object consisting of ice, dust and rocks is called:

- Asteroid
- ✓ Comet
- Planet
- Dwarf planet

(iv) Our sun is a:

- Planet
- Satellite
- ✓ Star
- Moon

(v) Moon is a:

- Satellite
- Planet
- Comet
- ✓ Natural satellite

10.2 Fill in the blanks.

(i) Our home planet is called ____.

Earth

(ii) In our solar system, the huge ball of burning gases that emit its own light is called ____.

Sun

(iii) Planets ____ around the Sun in their specific orbits.

revolve

(iv) The Moon does not have its own light, it ____ sunlight.

reflects

(v) A ____ eclipse happens when the Moon blocks the sun's light

from reaching the earth.

solar

(vi) Days and nights are caused by the ____ of the Earth about its own axis.

rotation

10.3 Match the column A with column B.

Match: Earth's axis, Lunar eclipse, New moon, The Sun, Revolution.

Earth's axis: Imaginary line Earth spins around. Lunar eclipse: Earth in between the Sun and the Moon. New moon: Looks dark or almost invisible. The Sun: Giant star at the center of solar system. Revolution: Planets circle around the Sun.

10.4 Write short answers.

(i) Define planetary system.

A planetary system is the group of planets and other heavenly bodies, such as comets, asteroids etc., revolving around the Sun. Our solar system is also an example of a planetary system.

(ii) Why do the shadows change their length?

Shadows form and keep on changing during the whole day, and the formation of shadows actually proves the rotation of the Earth. In the morning, as the Sun rises from the east, the Sun is low in the sky, so the shadows formed are long and stretched in the west direction; at noon the Sun comes exactly over our heads, so the shadows formed are shorter than the whole day; and after noon, as the Sun keeps on moving in the west, the shadows start to get

longer and stretch again in the east. Shadows are always formed in the opposite direction of the source of light, because light travels in straight lines and is blocked by opaque objects.

(iii) What is an asteroid?

Asteroids are the huge or small rocks floating in space and orbiting the Sun; they have an irregular shape and are not round like planets, and they can be as big as mountains and as small as pebbles.

(iv) What is the cause of the day and night on Earth?

Day and night are caused by the rotation of the Earth about its own axis: Earth completes one rotation about its own axis in 24 hours. In the morning, when the Sun rises, it gives us heat and light, and in the night when the Sun sets, darkness spreads everywhere, because the part of the Earth facing the Sun has day while the part away from the Sun has night.

10.5 Write detailed answers.

(i) Write a note on solar system.

Our solar system comprises the Sun, eight planets, moons, comets, asteroids etc., and Earth is the third nearest planet from the Sun. The center of our solar system is the Sun, and every other celestial body of the solar system revolves around the Sun. The eight planets are Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune, in order with respect to nearness to the Sun; the first four are called inner planets and have hard surfaces, while the last four are called the outer planets. Our solar system has only one star,

and that is the Sun, a very huge burning ball of gases and the biggest object in our solar system, placed in the center, providing all the heat and light to our Earth. The planet closest to the Sun is Mercury, and the planet closest to the Earth is Venus, and in our solar system Earth is the only planet that provides the conditions suitable for life.

(ii) Explain the phases of the moon.

The Moon revolves around the Earth, and we see the different shapes of the Moon; sometimes it is a thin sliver and sometimes we see the full moon. The Moon completes all its phases in 29.5 days. (i) New Moon: During this phase, the side of the Moon facing the Sun is illuminated while the side facing the Earth is not illuminated, so in this phase we cannot see the Moon at all in the sky. (ii) Crescent Moon: During this phase, the Moon looks like a bright thin sliver, emerging about one day after the new Moon, appearing as a faint, thin sliver of light in the sky after sunset. (iii) Half Moon: During this phase, the Moon exactly looks like a semi-circle, appearing when the Moon travels about a quarter of its way around the Earth, and up till this point half of the Moon is illuminated by sunlight. (iv) Full Moon: During this phase, the Moon looks like a complete bright circle, the brightest phase of the Moon, appearing once a month, approximately 15 days after a new moon.

10.6 PROJECT

Project (hands-on classroom task, not a written answer): Make a simple model of solar system. Place the Sun at the center and arrange the eight planets Mercury, Venus, Earth, Mars, Jupiter,

Saturn, Uranus and Neptune around it in their orbits, with the Moon revolving around the Earth.

Notes on the Book, Flags and Judgement Calls

Coverage: This PDF covers the COMPLETE book - all 10 units, every exercise (X.1 to X.6 in each unit). The book's Glossary (p.148) and Index (p.155) are word lists with no exercises, so they carry no solved items; they are noted here for completeness. Printed page numbers equal PDF page numbers in this book.

Flag 1 - 4.2(v): The book never uses the word 'xerophyte' in Unit 4 (pp.50-62); it calls cactus a desert plant ('Desert plants like cactus...'). The fill word used is therefore the book's own phrase, 'desert plant'. The standard science term for such a plant is xerophyte, but the book's wording was followed.

Flag 2 - 7.2(iii) and 7.2(v): Both resolved verbatim from Unit 7 lesson text - 'resistance' (section 7.2.3, p.102, and 'drag or resistance force', p.107) and 'efficiency' (section 7.3.2(i), p.105: 'Friction reduces the efficiency of machines').

Flag 3 - 9.2(i) and 9.2(v): 9.2(i) is 'naturally', from the book's own definition on p.125 ('materials found naturally on Earth'). 9.2(v) is 'soil': the unit never states the phrase 'topmost layer' verbatim, but soil is the unit's own match answer for 'in which plant roots grow', and it is the surface layer of the Earth.

Flag 4 - 6.3: The 'Loud sound - Lightning' pairing is the book's own column pairing, reproduced as printed. In the lesson text, thundering is listed as an example of a loud sound (p.89), and lightning is the flash that accompanies thunder, so the pairing follows the book's own columns.

Flag 5 - 10.1(v): The correct option is 'Natural satellite', not the plain 'Satellite' option, because the book states (p.138) that Earth has only one natural satellite, the Moon; artificial satellites are a different thing, so the plain 'Satellite' option is the distractor.

Flag 6 - 8.2(i): 'Technology makes our lives better' follows the book's introduction on p.113 ('to make the life better', 'making our lives better').

Judgement - Projects: Every unit's PROJECT (X.6) is recorded as a hands-on classroom task with a short guide, not as a solved question, because a project has no single written answer.

Judgement - Match exercises: Each X.3 match is given as one solved item listing all pairs, because the builder has no match item kind.

Judgement - 5.4(iv): The book asks why a needle sinks but does not give a one-line answer; the answer applies the book's own rule from p.75 (floating or sinking depends upon density; ice floats because it is less dense than water, coins sink because they are denser), so a metal needle sinks because its density is higher than water.

Book misprints noted, reproduced as printed: Unit 3, Fig. 3.6 labels a sparrow among the mammals (p.38); Unit 5, p.69 prints 'no indefinite shape and volume' for gases where the Key Points (p.81) correctly say 'no definite shape or volume'; Unit 7, p.107 prints 'maitains' for 'maintains'. None of these affect any exercise answer.